

Metro-Scan AI: Technical Algorithm & Architecture Specification

1. System Overview

Metro-Scan AI is an advanced computer vision compliance auditing system designed to verify multi-angle product packaging against **Rule 6 of the Indian Legal Metrology (Packaged Commodities) Rules, 2011**. The system ingests multi-image inputs representing different sides of a single product package, structures them into a unified multimodal payload, queries a high-performance Vision-Language Model (VLM) via Groq, strips internal reasoning tokens, and produces a structured audit report detailing explicit [PASS] or [FAIL] compliance statuses alongside required corrective actions.

2. Input Acquisition & State Management

- **Mode Selection:** The application uses Streamlit session state (`st.session_state.input_mode`) to toggle between two distinct input mechanisms:
 1. *Bulk File Uploader:* `st.file_uploader` accepting standard image formats (JPG, JPEG, PNG) with multi-file support.
 2. *Live Camera Capture:* Progressive live snapshots captured via `st.camera_input`, appending unique frames sequentially to `st.session_state.live_photos` while filtering out duplicate consecutive captures.
- **Pipeline Normalization:** Active images from either input mode are consolidated into a unified `uploaded_images` array.

3. Multimodal Preprocessing & Payload Construction

For every image present in the active session array, the preprocessing pipeline executes the following sequence:

1. **Byte Extraction:** Read the raw byte buffer from the uploaded file object (`img.getvalue()`).
2. **Base64 Encoding:** Convert the raw binary stream into a UTF-8 decoded Base64 string (`base64.b64encode`).
3. **Payload Structuring:** Format the request payload for the Groq API client:
 - **Text Component:** Contains strict audit instructions and evaluation criteria emphasizing multi-image aggregation (treating all uploaded

images as different sides of the *exact same* package).

- **Image Components:** Array of multimodal image objects embedding Base64 data URLs (data:image/jpeg;base64,...).

4. Inference & VLM Execution Algorithm

- **Model Dispatch:** Sends the constructed multimodal payload to the qwen/qwen3.6-27b model via the Groq SDK.
- **Hyperparameter Configuration:**
 - temperature = 0.1: Minimized to enforce deterministic, analytical consistency and eliminate generative hallucinations.
 - max_tokens = 800: Configured to operate within Groq's free-tier output tokens per minute (OTPM) rate limits.
- **Prompt Engineering Constraints:**
 - Enforces cross-image context (combining attributes found across any uploaded side).
 - Enforces strict validation rules (e.g., rejecting vegetarian green dots as Country of Origin, requiring explicit "MRP" labeling, and treating "Lorem ipsum" placeholders as failures).

5. Output Sanitization & Parsing Algorithm

To remove internal model chain-of-thought tokens (<think> blocks) and ensure a clean user interface:

1. **Regex Substitution:** Scan the raw model response string (raw_output) and strip all text encapsulated inside reasoning tags using regular expression pattern matching (re.sub(r'<think>.*?</think>', "", raw_output, flags=re.DOTALL)).
2. **Fallback Isolation:** In edge cases where token limits result in unclosed tags, execute a secondary safety check to isolate valid output text.
3. **Dynamic Rendering:** Display the sanitized audit report using markdown formatting within the Streamlit application interface.